

Jenna Metera

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Education

University of California, San Diego, Jacobs School of Engineering, La Jolla, California

Master of Science, Materials Science and Engineering, December 2019

Doctorate of Philosophy 5th Year Ph.D. Student (Projected: December 2024)

Major: Materials Science and Engineering

Alfred University, Kazuo Inamori School of Engineering, Alfred, New York

Bachelor of Science May 2018 GPA: 3.43

Major: Ceramic Engineering **Minors:** Business Administration, Chemistry

Graduate Courses Completed: Intro to X-ray Crystallography, Marketing, Business Administration, Finance, Intro to NanoEngineering, Nanosystems Integration, Electronic & Photon Properties of Materials, Phase Transformations, Nanomaterials & Properties, Energy Materials & Applications, Thermodynamics of Solids, Solid State Diffusion & Reaction Kinetics, Imperfections in Solids, Structure & Analysis of Solids

Skills & Equipment

Skills: Synthesis Tuning, AS9100 Certified, ISO9001 Certified, 5S, PFMEA and CP development, Capital Equipment Acquisition, Electrical Engineering, Sustainability, Electronic Planning, Liquid Tin Process Engineering, Materials Synthesis, Manufacturing Process Improvement, Material Design, Time Management, Semiconductor Process Engineering, Communication, Applications Engineering, Research, Teamwork, Milling Studies, Experimental Design, Supply Chain Management, Material Characterization, Electrical Material Characterization, Chemical Engineering, Mud Lab (batching, filter pressing, extruding, firing), Powder Processing, Suspension Rheology, Tape Casting, Firing Studies, Furnace Optimization, Customer Service, Polymer Rheology, Technical Writing, Batch Calculations, Teaching, Microsoft (MS Office), RMA/MRB/PCRB familiarity, Yield Analysis

Equipment and Programs: XRD, SEM, FIB, TEM, Microhardness Testing, Dorst Press, DISCO Saws, Belt Furnace, Stationary Furnace, Tristar II, Electrical Characterization Equipment, Sedigraph III, Brookfield Viscometer, Helium Pycnometer, Dynamic Shear Rheometer, Adobe Photoshop, Microsoft Suite

Presentations and Posters

Jenna Metera and Olivia A. Graeve, “Morphologically cubic BiFeO₃ for improved electrical properties”. *The American Ceramic Society*. Electronic Materials and Applications Conference. January 2021.

Jenna Metera and Olivia A. Graeve. “Cubic BiFeO₃ powders for improved electrical properties”. *The Minerals, Metals, & Materials Society*. February 2019. Presentation.

Jenna Metera and Steven M. Pilgrim. “Dielectric response uncoupled from strain tensor in relaxor ceramics”. *State University of New York Undergraduate Research Conference*. April 2018. Poster.

Jenna Metera and Steven M. Pilgrim. “Dielectric response uncoupled from strain tensor in relaxor ceramics”. *Alfred University Undergraduate Research Conference*. April 2018. Poster.

Engineering and Research Work Experience

Integrated Microwave Corp. – Knowles Precision Devices San Diego, California

Ceramic Process Engineer and Capital Equipment Asset Manager (November 2021 – Present)

- Lead the search for formation technique to eliminate delaminations in tape cast ceramic at Cazenovia, NY site and successfully fostered relationship with US-CEO of Dorst America to rapidly test and acquire vital press equipment
- Established the Dicing Center of Excellence using cross facility relationships to optimize the quality of diced products across the Precision Devices business sector, increasing yield by 20% and OTD across the sector
- Increased yield of ceramic dielectric substrates by 30%. Original substrates resulted in lamellar cracking where study of formation pressure, firing rate, flatness firing, and dicing conditions led to a further optimized product
- Developed product specific resonator marketing tools to attract new customers using multiple channels. Incorporated competitive pricing leveraging the excellent quality with competitors’ products
- Implemented and lead new process for Capital Acquisition Report (CAR) development for CAPEX projects catering to our site’s specific needs including technical and financial justification
- Conducted the implementation for PFMEA and CP analysis across the San Diego site

ASML San Diego, California

Ceramic Coating Research Assistant (August 2019 - Present)

- Identified several chemically corrosive resistant materials through intensive literature search
- Designed experiments to characterize the performance and success of each material

EUV Test Capability Engineer Intern (Summer 2021)

- Examined current and proposed testing capability in San Diego and Veldhoven (the Netherlands)
- Identified tests completed and future tests that should be completed and how they align with current roadmap needs
- Identified gaps in the current and future test capabilities and recommended solutions to get rid of gaps

EUV Materials Engineer Intern (Summer 2020)

- Organized data on the EUV system to improve semiconductor processing time
- Communicated with the management team of liquid tin problem areas
- Proposed new material solutions to mitigate problem areas

Owens Corning Granville, Ohio

Science and Technology Roofing and Asphalt Intern (Summer 2018)

Asphalt Innovation using Rheology:

- Characterized polymer modified asphalts with dynamic shear rheology for sealants, coatings, adhesives in roofing
- Examined novel rheological results to compare to competitors. Recommended new formulations based on data

Knowles Capacitors Norwich, Norfolk UK & Cazenovia, New York USA

Ceramic Engineering Production Intern (Summer 2017)

Ceramic Plate Improvement Project:

- Examined raw materials' chemical formulations to learn why they were variable by 20% in the casting process
- Determined the current tape thickness method was too variable and recommended a less variable system
- Defined methods to improve viscosity in batching/casting which had benefits like easier casting and less waste
- Reduced the variation in relative permittivity by ± 0.1 by batching more consistently

Ceramic Engineering Intern (Summer 2016)

- Minimized mill time and maximized batch size by experimenting with several amounts of media/batch
- Recommended further study of mill jar safety. Collected data on mixed multi-batch containers to reduce variance
- Developed an abrasion technique, maintained tube furnaces, and tested the final product
- Observed processing methods to gain knowledge on R&D process and offered new hypotheses if experiments failed

Awards and Publications

- NSF Internship Award, ASML Research, January 2023
- NSF DMR Award, Graduate Research, September 2022
- Abe Hurlich Scholarship, ASM San Diego Chapter, May 2020
- Dr. David R. Jones IV Scholarship, Association of Modified Asphalt Producers, January 2019
- Jacobs School of Engineering Fellowship, September 2018
- Sigma Xi "Outstanding Undergraduate Research Award", April 2018
- Dean's List (Alfred University) - Fall 2014, Fall 2018, Spring 2018

Activities

Chair, ASM San Diego, April 2022 - Present

Mentor, SDSU Women in Engineering Program, March 2021 - Present

Mentor, Jacobs School Mentoring Program (Grad Pals, JUMP), September 2020 - Present

Member, American Ceramic Society, December 2019 - Present

Member, European Ceramic Society, December 2019 - Present

Executive Board Member, ASM San Diego, November 2019 - Present

Mentor, MAE Women's Club UC San Diego, September 2018 - Present

Member, Keramos, April 2016 - Present

Member, Materials Advantage, August 2015 - May 2018

Member, Society of Women's Engineers (SWE), August 2014 - Present

Treasurer, Keramos, April 2016 - January 2017

Activities Organizer, AU Bowling Club, December 2014 - September 2016

Media Manager & Secretary, AUTV, August 2014 - August 2015